

# Proposal 2: Tensor greedy double extended Kaczmarz with t-product for solving large-scale inconsistent linear tensor systems

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# Outline

# Abstract

Tensor are intensively used to represent images, videos, and any higher dimensional object. In image reconstruction, given a blurred and noisy images, the reconstruction of that image can be expressed as a tensor linear systems. In this note, we present a tensor greedy double extended Kaczmarz for solving large scale tensor linear systems. Unlike the block-based methods TREBK, TREGKB, TREABK, the proposed method TGDBEK **does not require any pre-defined partitions of the tensor systems**. It employs the greedy principle on the tensor systems to dynamically construct a relatively small subset of rows and columns. Then updates the iterates using a REK update style. We provide the convergence guarantees of the proposed method. We have shown that the method convergence exponentially in expectation to the minimum-norm solutions. Numerical examples are conducted to assess the effectiveness of the proposed method.

# Notations and preliminaries

- ▶ A tensor is a natural generalization of a matrix in higher dimension. See fig. ??.
- ▶ We borrow our notations from [?, ?]. We denote tensors by calligraphic letters  $\mathcal{A}$ .
- ▶ Throughout this note, we consider third-order tensors, i.e.  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$
- ▶ Given a tensor  $\mathcal{A}$ , let  $\mathbf{A}_i := \mathcal{A}_{:, :, i}$ . The block circulant matrix of  $\mathcal{A}$  is defined as [?]:

$$\text{bcirc}(\mathcal{A}) := \begin{bmatrix} \mathbf{A}_1 & \mathbf{A}_{N_3} & \cdots & \mathbf{A}_2 \\ \mathbf{A}_2 & \mathbf{A}_1 & \cdots & \mathbf{A}_3 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{A}_{N_3} & \mathbf{A}_{N_3-1} & \cdots & \mathbf{A}_1 \end{bmatrix} \in \mathbb{R}^{N_1 N_3 \times N_2 N_3}.$$

slides-images/3d-tensor.jpg

slides-images/3d-tensor-2.png

Figure: Illustration of third-order tensors.

- We define the operations  $\text{fold}(\cdot)$  and its inverse  $\text{unfold}(\cdot)$  as follows:

$$\text{unfold}(\mathcal{A}) := \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_{N_3} \end{bmatrix} \in \mathbb{R}^{N_1 N_3 \times N_2}, \quad \text{fold} \left( \begin{bmatrix} \mathbf{A}_1 \\ \mathbf{A}_2 \\ \vdots \\ \mathbf{A}_{N_3} \end{bmatrix} \right) := \mathcal{A}.$$

- In [?], the authors proposed a new tensor to tensor products (named t-product), which allows one to extend matrix linear algebra to tensors. The following is the definition of the t-product.

### Definition (1, [?])

For  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$  and  $\mathcal{B} \in \mathbb{R}^{N_2 \times K \times N_3}$ , the *t-product* between  $\mathcal{A}$  and  $\mathcal{B}$  is a  $N_1 \times K \times N_3$  tensor defined and denoted  $\mathcal{A} * \mathcal{B}$

$$\mathcal{A} * \mathcal{B} := \text{fold}(\text{bcirc}(\mathcal{A})\text{unfold}(\mathcal{B})) \in \mathbb{R}^{N_1 \times K \times N_3}$$

- ▶ The product  $\text{bcirc}(\mathcal{A})\text{unfold}(\mathcal{B})$  is a usual matrix product.
- ▶ The resulting matrix is a block matrix which then is folded to a tensor.

We summarize below some definitions and properties that we will use in this note.

### Definition ([?, ?, ?])

Let  $\mathcal{A} \in \mathbb{R}^{N_1 \times \times N_2 \times N_3}$  and  $\mathcal{B} \in \mathbb{R}^{N_2 \times K \times N_3}$ . Then the t-product between  $\mathcal{A}$  and  $\mathcal{B}$  satisfy the following properties

1. The separability in the first dimension, that is

$$(\mathcal{A} * \mathcal{B})_{i, :, :} = \mathcal{A}_{i, :, :} * \mathcal{B}.$$

2. The component form decomposition on the second dimension

$$\mathcal{A} * \mathcal{B} = \sum_{j=1}^{N_2} \mathcal{A}_{:, j, :} * \mathcal{B}_{j, :, :}. \quad (1)$$

### Definition (transpose)

The transpose of  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ , denoted by  $\mathcal{A}^\top$ , is the  $N_2 \times N_1 \times N_3$  tensor obtained by transposing each of the frontal slices and then reversing the order of transposed frontal slices 2 through  $N_3$ . For  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ , we have

$$\text{bcirc}(\mathcal{A}^\top) = (\text{bcirc}(\mathcal{A}))^\top. \quad (2)$$

### Definition (inner product)

The inner product between  $\mathcal{A}$  and  $\mathcal{B}$  in  $\mathbb{R}^{N_1 \times N_2 \times N_3}$  is defined as

$$\mathcal{A}, \mathcal{B} := \sum_{i,j,k} \mathcal{A}_{ijk} \mathcal{B}_{ijk}.$$

## Definition

For  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ , its spectral norm and the Frobenius norm are defined as follows

$$\|\mathcal{A}\|_2 := \|\text{bcirc}(\mathcal{A})\|_2 \quad (3)$$

$$\|\mathcal{A}\|_F := \sqrt{\langle \mathcal{A}, \mathcal{A} \rangle} = \sqrt{\sum_{i,j,k} \mathcal{A}(i,j,k)^2} \quad (4)$$

## Lemma ([?])

For  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$  and  $\mathcal{B} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ , it holds that

$$\|\mathcal{A} * \mathcal{B}\|_F \leq \|\mathcal{A}\|_2 \|\mathcal{B}\|_F \quad (5)$$

### Definition

For  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ , its pseudoinverse, denoted by  $\mathcal{A}^\dagger$ , is the  $N_2 \times N_1 \times N_3$  tensor satisfying

$$\text{bcirc}(\mathcal{A}^\dagger) = \text{bcirc}(\mathcal{A})^\dagger \quad (6)$$

### Lemma

For  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$  and  $\mathcal{B} \in \mathbb{R}^{N_2 \times K \times N_3}$ , the following holds

$$\mathcal{A}^\top * \mathcal{A} * \mathcal{A}^\dagger * \mathcal{B} = \mathcal{A}^\top * \mathcal{B} \quad (7)$$

In fact, we know that  $\mathcal{A} * \mathcal{A}^\top$  is symmetric,  $\mathcal{A} * \mathcal{A}^\dagger * \mathcal{A} = \mathcal{A}$  and  $(\mathcal{A}^\dagger)^\top = (\mathcal{A}^\top)^\dagger$ . We have

$$\mathcal{A}^\top * \mathcal{A} * \mathcal{A}^\dagger * \mathcal{B} = \mathcal{A}^\top * (\mathcal{A}^\dagger)^\top * \mathcal{A}^\top * \mathcal{B} = \mathcal{A}^\top * (\mathcal{A}^\top)^\dagger * \mathcal{A} = \mathcal{A}^\top * \mathcal{B}$$

### Lemma

For  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$  and  $\mathcal{X} \in \text{range}(\mathcal{A})$ , the following holds

$$\|\mathcal{A}^\top * \mathcal{X}\|_F^2 \geq \sigma_{\min}^2(\text{bcirc}(\mathcal{A})) \|\mathcal{X}\|_F^2 \quad (8)$$

*With the algebra defined above, it becomes clear that the t-product is for tensors what the product (composition) is for matrices.*

# Motivation

- ▶ We aim to solve the linear systems

$$\mathcal{A} * \mathcal{X} = \mathcal{B}, \mathcal{B} = \overline{\mathcal{B}} + \epsilon \quad (9)$$

where  $\epsilon$  is a noisy, and  $\overline{\mathcal{B}}$  is the clean right-hand side.

- ▶ This problem arises in image reconstruction [?, ?].
- ▶ To solve this problem, [?] proposed block-based tensor randomized extended Kaczmarz, namely TREBK, TREGKB.
- ▶ The TREGKB uses a greedy strategies which discriminates blocks and projects the current iterate on a set of block whose Frobenius norm is bigger than some threshold.

- ▶ In the same line, Ann et al. [?] adapted recently the randomized extended average Kaczmarz (REABK) [?] to the tensor framework under the t-product to obtain the tensor randomized extended average Kaczmarz (TREABK).
- ▶ Unlike the TREGKB or TREBK, the TREABK doesn't project the current iterates on the block hyperplanes forming the selected block. But, it uses a weighted convex combination of projection the the current iterate on each selected hyperplane.

- ▶ The averaging idea seems to ameliorate the convergence of the TREABK compare to TREBK and TREGKB.
- ▶ What the methods have in common is the **pre-partition** of the tensor systems: they require pre-defined blocks sub-tensors to perform.
- ▶ One of the advantageous of using a greedy technique is it specifies a discrimination of criterion on hyperplanes, which often results to less running time and less iteration. Hence, a better convergence decay  $[?, ?]$  compared to non-greedy methods.

# Tensor Greedy Double Block Extended Kaczmarz (TGDBEK)

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**Algorithm 1:** Tensor Greedy Double Block Extended Kaczmarz (TGDBEK)

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**Input:**  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ ,  $\mathcal{B} \in \mathbb{R}^{N_1 \times K \times N_3}$ , iters  $\ell$ ,  $\eta \in (0, 1]$ .

**Init:**  $\mathcal{X}^{(0)} = \mathcal{O}$ ,  $\mathcal{Z}^{(0)} = \mathcal{B}$ .

**for**  $k = 0, 1, \dots, \ell - 1$  **do**

$$\varepsilon_k^z = \eta \max_{j \in [N_2]} \frac{\|(\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^{(k)}\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2}$$

$$U_k = \left\{ j \in [N_2] : \|(\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^{(k)}\|_F^2 \geq \varepsilon_k^z \|\mathcal{A}_{:,j,:}\|_F^2 \right\}$$

$$\mathcal{Z}^{(k+1)} = \mathcal{Z}^{(k)} - \mathcal{A}_{:,U_k,:} * (\mathcal{A}_{:,U_k,:})^\dagger * \mathcal{Z}^{(k)}$$

$$\varepsilon_k^x = \eta \max_{i \in [N_1]} \frac{\|\mathcal{B}_{i,:,:} - \mathcal{Z}_{i,:,:}^{(k+1)} - \mathcal{A}_{i,:,:} * \mathcal{X}^{(k)}\|_F^2}{\|\mathcal{A}_{i,:,:}\|_F^2}$$

$$J_k = \left\{ i \in [N_1] : \|\mathcal{B}_{i,:,:} - \mathcal{Z}_{i,:,:}^{(k+1)} - \mathcal{A}_{i,:,:} * \mathcal{X}^{(k)}\|_F^2 \geq \varepsilon_k^x \|\mathcal{A}_{i,:,:}\|_F^2 \right\}$$

$$\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} + (\mathcal{A}_{J_k,:,:})^\dagger * \left( \mathcal{B}_{J_k,:,:} - \mathcal{Z}_{J_k,:,:}^{(k+1)} - \mathcal{A}_{J_k,:,:} * \mathcal{X}^{(k)} \right)$$

**end**

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- ▶ The TGDEBK uses the whole tensors systems.
- ▶ It maintains the same update style as REK. In fact,
- ▶ Given a primal iterate  $\mathcal{X}^{(k)}$ , it selects avariciously the slices whose distance between  $\mathcal{X}^{(k)}$  are relatively greater than some proportional of the max distance of all the distance of  $\mathcal{X}^{(k)}$  between all the systems indexes.
- ▶ The GDBEK was introduced in [?].

## Theorem

Assume the system (??) is inconsistent and let  $\mathcal{B}^\perp := \mathcal{B} - \mathcal{A}\mathcal{A}^\dagger\mathcal{B}$ . The tensor sequence  $\mathcal{X}^{(k)}$  generated by TGDBEK algorithm converges to the tensor least-square solution  $\mathcal{X}_* = \mathcal{A}^\dagger * \mathcal{B}$ . Moreover, for  $k \geq 1$ , the following holds

$$\|\mathcal{X}^{(k+1)} - \mathcal{X}_*\|_F^2 \leq \alpha^{k+1} \|\mathcal{X}_0 - \mathcal{X}_*\|_F^2 + \frac{\alpha^{k+1} - \beta^{k+1}}{\alpha - \beta} \|\mathcal{B}^\perp\|_F^2 \quad (10)$$

where  $\alpha = 1 - \theta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}^\top))}{\sigma_{\max}^2(\text{bcirc}(\mathcal{A}^\top))}$ ,  $\beta = 1 - \frac{\sigma_{\max}^2(\mathcal{A})}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}(\overline{U})\|_F^2}$ ,  $\gamma = \sigma_{\min}^2(\mathcal{A}^\mathcal{J})$ ,  $\theta \in (0, 1]$  and  $\eta \in (0, 1]$ .

## Proof.

From Lemma 5, we obtain

$$\mathcal{A}^\top * \mathcal{B}^\perp = \mathcal{A}^\top * \mathcal{B} - \mathcal{A}^\top * \mathcal{A} * \mathcal{A}^\dagger * \mathcal{B} = \mathcal{A}^\top * \mathcal{B} - \mathcal{A}^\top * \mathcal{B} = 0$$

In particular, for any  $U_k \subset [N_2]$  we get  $\mathcal{A}(:, U_k, :)^\top \mathcal{B}^\perp = 0$ . Now, using the fact  $\mathcal{A} * \mathcal{A}^\dagger = (\mathcal{A} * \mathcal{A}^\dagger)^\top = (\mathcal{A}^\dagger)^\top * \mathcal{A}^\top = (\mathcal{A}^\top)^\dagger * \mathcal{A}^\top$  and the z-update in Alg 1, we can get

$$\begin{aligned} \mathcal{Z}^{(k+1)} - \mathcal{B}^\perp &= (\mathcal{Z}^{(k)} - \mathcal{B}^\perp) - (\mathcal{A}_{:,U_k,:}^\top)^\dagger * \mathcal{A}_{:,U_k,:}^\top * (\mathcal{Z}^{(k)} - \mathcal{B}^\perp) \\ &= \left( \mathcal{I}_{:,U_k,:} - (\mathcal{A}_{:,U_k,:}^\top)^\dagger * \mathcal{A}_{:,U_k,:}^\top \right) * (\mathcal{Z}^{(k)} - \mathcal{B}^\perp). \end{aligned}$$

Since  $(\mathcal{A}_{:,U_k,:}^\top)^\dagger * \mathcal{A}_{:,U_k,:}^\top$  is a  $t$ -tensor projector, using the Pythagorean theorem, we can obtain



$$\|\mathcal{Z}^{(k+1)} - \mathcal{B}^\perp\|_F^2 = \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2 - \left\| \left( \mathcal{A}_{:,U_k,:}^\top \right)^\dagger * \mathcal{A}_{:,U_k,:}^\top * \left( \mathcal{Z}^{(k)} - \mathcal{B}^\perp \right) \right\|_F^2. \quad (11)$$

Using Lemma (), we note that

$$\begin{aligned} \left\| \left( \mathcal{A}_{:,U_k,:}^\top \right)^\dagger * \mathcal{A}_{:,U_k,:}^\top * \left( \mathcal{Z}^{(k)} - \mathcal{B}^\perp \right) \right\|_F^2 &= \left\| \left( \mathcal{A}_{:,U_k,:}^\top \right)^\dagger * \mathcal{A}_{:,U_k,:}^\top * \mathcal{Z}^{(k)} \right\|_F^2 \\ &\geq \sigma_{\min}^2 \left( \left( \mathcal{A}_{:,U_k,:}^\top \right)^\dagger \right) \left\| \mathcal{A}_{:,U_k,:}^\top * \mathcal{Z}^{(k)} \right\|_F^2 \\ &= \sigma_{\min}^2 (\text{bcirc}(\mathcal{A}_{:,U_k,:}^\top)^\dagger) \sum_{j \in U_k} \left\| \mathcal{A}_{:,j,:}^\top * \mathcal{Z}^{(k)} \right\|_F^2 \\ &\geq \sigma_{\min}^2 (\text{bcirc}(\mathcal{A}_{:,U_k,:}^\top)^\dagger) \epsilon_k^z \left\| \mathcal{A}_{:,U_k,:}^\top \right\|_F^2 \\ &= \frac{\left\| \mathcal{A}_{:,U_k,:}^\top \right\|_F^2}{\sigma_{\max}^2 (\text{bcirc}(\mathcal{A}_{:,U_k,:}^\top))} \epsilon_k^z. \end{aligned} \quad (12)$$

Now, using the lemma (), we observe that

$$\begin{aligned} \mathcal{A}_{:,U_{k-1},:}^\top \mathcal{Z}^{(k)} &= \mathcal{A}_{:,U_{k-1},:}^\top \left( \mathcal{Z}^{(k-1)} - \mathcal{A}_{:,U_{k-1},:} * \mathcal{A}_{:,U_{k-1},:}^\dagger \mathcal{Z}^{k-1} \right) \\ &= \mathcal{A}_{:,U_{k-1},:}^\top * \mathcal{Z}^{(k-1)} - \mathcal{A}_{:,U_{k-1},:}^\top * \mathcal{A}_{:,U_{k-1},:} * \mathcal{A}_{:,U_{k-1},:}^\dagger * \mathcal{Z}^{k-1} = \mathcal{O}_z \end{aligned}$$

so that for all  $k = 0, 1, \dots$ , we can get

$$\begin{aligned}
\|\mathcal{A}^\top \mathcal{Z}^{(k)}\|_F^2 &= \sum_{j \in U_k} \|\mathcal{A}_{:,j,:}^\top * \mathcal{Z}^{(k)}\|_F^2 + \sum_{j \in U_{k-1}^c} \|\mathcal{A}_{:,j,:}^\top * \mathcal{Z}^{(k)}\|_F^2 \\
&= \sum_{j \in U_{k-1}^c} \|\mathcal{A}_{:,j,:}^\top * \mathcal{Z}^{(k)}\|_F^2 = \sum_{j \in U_{k-1}^c} \frac{\|\mathcal{A}_{:,j,:}^\top * \mathcal{Z}^{(k)}\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2} \|\mathcal{A}_{:,j,:}\|_F^2 \\
&\leq \max_{j \in [N_2]} \left\{ \frac{\|\mathcal{A}_{:,j,:}^\top * \mathcal{Z}^{(k)}\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2} \right\} \left( \|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:,U_k,:}\|_F^2 \right).
\end{aligned}$$

Hence

$$\begin{aligned}
\epsilon_k^z &= \eta \max_{j \in [N_2]} \left\{ \frac{\|\mathcal{A}_{:,j,:}^\top * \mathcal{Z}^{(k)}\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2} \right\} \geq \eta \frac{\|\mathcal{A}^\top \mathcal{Z}^{(k)}\|_F^2}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:,U_k,:}\|_F^2} \\
&\geq \eta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}^\top)) \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:,U_k,:}\|_F^2}, \tag{13}
\end{aligned}$$

where in the last inequality we used

$$\|\mathcal{A}^\top * \mathcal{Z}^{(k)}\|_F^2 = \|\mathcal{A}^\top * (\mathcal{Z}^{(k)} - \mathcal{B}^\perp)\|_F^2 \geq \sigma_{\min}^2(\text{bcirc}(\mathcal{A}^\top)) \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2.$$

Substitute (??) in (??) we can obtain

$$\left\| \left( \mathcal{A}_{:,u_k,:}^\top \right)^\dagger * \mathcal{A}_{:,u_k,:}^\top * \left( \mathcal{Z}^{(k)} - \mathcal{B}^\perp \right) \right\|_F^2 \geq \frac{\eta \|\mathcal{A}_{:,u_k,:}\|_F^2}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:,u_k,:}\|_F^2} \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}^\top)) \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2}{\sigma_{\max}^2(\text{bcirc}(\mathcal{A}_{:,u_k,:}^\top))}. \quad (14)$$

Let  $\|\mathcal{A}_{:,\bar{U},:}\|_F^2 := \max_k \{\|\mathcal{A}_{:,u_k,:}\|_F^2\}$  so that, combining (??) and (??) above, we can finally obtain the upper bound

$$\begin{aligned} \|\mathcal{Z}^{(k+1)} - \mathcal{B}^\perp\|_F^2 &\leq \left( 1 - \frac{\eta \|\mathcal{A}_{:,u_k,:}\|_F^2}{\sigma_{\max}^2(\text{bcirc}(\mathcal{A}_{:,u_k,:}^\top)) \|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:,u_k,:}\|_F^2} \right) \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2 \\ &\leq \left( 1 - \eta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}^\top))}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:,\bar{U},:}\|_F^2} \right) \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2. \end{aligned} \quad (15)$$

where in the last inequality we used the fact that  $\frac{\|\mathcal{A}_{:,u_k,:}\|_F^2}{\sigma_{\max}^2(\text{bcirc}(\mathcal{A}_{:,u_k,:}^\top))} \geq 1$ . The fact that we can decompose  $\mathcal{B}_{J_k,:} = \mathcal{A}_{J_k,:} * \mathcal{X}^* + \mathcal{B}_{J_k,:}^\perp$ , we can rewrite the the  $x$ - update as,

$$\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} - \mathcal{A}_{J_k,:}^\dagger * \left( \mathcal{A}_{:,J_k,:} * \left( \mathcal{X}^{(k)} - \mathcal{X}^* \right) + \left( \mathcal{Z}_{J_k,:}^{(k+1)} - \mathcal{B}_{J_k,:}^\perp \right) \right)$$

Subtracting  $\mathcal{X}_*$  both sides in the equality above, we obtain

$$\begin{aligned}\mathcal{X}^{(k+1)} - \mathcal{X}_* &= \mathcal{X}^{(k)} - \mathcal{X}_* - \mathcal{A}_{J_k, :, :}^\dagger * \left( \mathcal{A}_{:, J_k, :} * (\mathcal{X}^{(k)} - \mathcal{X}^*) + \left( \mathcal{Z}_{J_k, :, :}^{(k+1)} - \mathcal{B}_{J_k, :, :}^\perp \right) \right) \\ &= \left( \mathcal{I} - \mathcal{A}_{J_k, :, :}^\dagger * \mathcal{A}_{:, J_k, :} \right) * (\mathcal{X}^{(k)} - \mathcal{X}^*) + \mathcal{A}_{J_k, :, :}^\dagger * \left( \mathcal{Z}_{J_k, :, :}^{(k+1)} - \mathcal{B}_{J_k, :, :}^\perp \right)\end{aligned}\quad (16)$$

Again  $\mathcal{A}_{:, J_k, :}^\dagger \mathcal{A}_{:, J_k, :}$  is a  $t$ -orthogonal projector (slice-wise orthogonal in the Fourier domain), hence the two terms are orthogonal in the Frobenius norm. Therefore, from the Pythagorean theorem, we can obtain

$$\|\mathcal{X}^{(k+1)} - \mathcal{X}_*\|_F^2 = \left\| \left( \mathcal{I} - \mathcal{A}_{J_k, :, :}^\dagger * \mathcal{A}_{:, J_k, :} \right) * (\mathcal{X}^{(k)} - \mathcal{X}^*) \right\|_F^2 + \left\| \mathcal{A}_{J_k, :, :}^\dagger * \left( \mathcal{Z}_{J_k, :, :}^{(k+1)} - \mathcal{B}_{J_k, :, :}^\perp \right) \right\|_F^2 \quad (17)$$

Let  $\mathcal{X}^{(k)} - \mathcal{X}_* = \left( \mathcal{X}^{(k)} - \mathcal{X}_* \right)_{\mathcal{R}((\mathcal{A}_{:, J_k, :})^\top)} + \left( \mathcal{X}^{(k)} - \mathcal{X}_* \right)_{\mathcal{R}((\mathcal{A}_{:, J_k, :})^\top)^\perp}$  and

$\left\| \left( \mathcal{X}^{(k)} - \mathcal{X}_* \right)_{\mathcal{R}((\mathcal{A}_{:, J_k, :})^\top)} \right\|_F^2 = \theta_k \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2$ . Then, using the Pythagorean theorem and the Lemma (), we get :

$$\begin{aligned}
\| (\mathcal{I} - \mathcal{A}_{J_k, :, :}^\dagger * \mathcal{A}_{:, J_k, :}) * (\mathcal{X}^{(k)} - \mathcal{X}^*) \|_F^2 &= \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 - \|\mathcal{A}_{J_k, :, :}^\dagger * \mathcal{A}_{:, J_k, :} * (\mathcal{X}^{(k)} - \mathcal{X}^*) \|_F^2 \\
&\leq \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 \\
&\quad - \sigma_{\min}^2(\mathcal{A}_{:, J_k, :}^\dagger) \sigma_{\min}^2(\mathcal{A}_{:, J_k, :}) \| (\mathcal{X}^{(k)} - \mathcal{X}_*)_{R((\mathcal{A}_{:, J_k, :})^\top)} \|_F^2 \\
&= \left( 1 - \theta_k \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}_{:, J_k, :}))}{\sigma_{\max}^2(\mathcal{A}_{:, J_k, :})} \right) \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 \\
&\leq \left( 1 - \theta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}_{:, J, :}))}{\sigma_{\max}^2(\mathcal{A}_{:, J, :})} \right) \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2, \tag{18}
\end{aligned}$$

where we denote  $J = \{J_0, 1, \dots, J_k\}$  and  $\theta := \min_k \{\theta_k\}$ , and  $\mathcal{A}_{:, J, :}$  the stacked or the union block of tensors  $\mathcal{A}_{:, J_k, :}$ . For the second term, we first note that  $\|\mathcal{A}^\dagger \mathcal{X}\|_F \leq \|\mathcal{A}^\dagger\|_2^2 \|\mathcal{X}\|_F^2$ .

Then, we obtain

$$\begin{aligned}
\|\mathcal{A}_{J_k, :, :}^\dagger * (\mathcal{Z}_{J_k, :, :}^{(k+1)} - \mathcal{B}_{J_k, :, :}^\perp)\|_F^2 &\leq \sigma_{\max}^2((\mathcal{A}_{:, J_k, :})^\dagger) \|\mathcal{Z}_{J_k, :, :}^{(k+1)} - \mathcal{B}_{J_k, :, :}^\perp\|_F^2 \\
&= \frac{1}{\sigma_{\min}^2(\mathcal{A}_{:, J_k, :})} \|\mathcal{Z}_{J_k, :, :}^{(k+1)} - \mathcal{B}_{J_k, :, :}^\perp\|_F^2.
\end{aligned} \tag{19}$$

It results from (??), (??), and (??) that

$$\begin{aligned}
&\|\mathcal{X}^{(k+1)} - \mathcal{X}_*\|_F^2 \\
&\leq \left(1 - \theta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}_{:, J, :}))}{\sigma_{\max}^2(\mathcal{A}_{:, J, :})}\right) \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 + \frac{1}{\sigma_{\min}^2(\mathcal{A}_{:, J_k, :})} \|\mathcal{Z}_{J_k, :, :}^{(k+1)} - \mathcal{B}_{J_k, :, :}^\perp\|_F^2 \\
&\leq \left(1 - \theta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}_{:, J, :}))}{\sigma_{\max}^2(\mathcal{A}_{:, J, :})}\right) \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 + \frac{1}{\sigma_{\min}^2(\mathcal{A}_{:, J_k, :})} \left(1 - \eta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}^\top))}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:, \bar{U}, :}\|_F^2}\right) \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2.
\end{aligned}$$

It results from (??), (??), and (??) that

$$\begin{aligned}
& \|\mathcal{X}^{(k+1)} - \mathcal{X}_*\|_F^2 \\
& \leq \left(1 - \theta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}_{:,J,:}))}{\sigma_{\max}^2(\mathcal{A}_{:,J,:})}\right) \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 + \frac{1}{\sigma_{\min}^2(\mathcal{A}_{:,J_k,:})} \|\mathcal{Z}_{J_k,:}^{(k+1)} - \mathcal{B}_{J_k,:}^\perp\|_F^2 \\
& \leq \left(1 - \theta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}_{:,J,:}))}{\sigma_{\max}^2(\mathcal{A}_{:,J,:})}\right) \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 + \frac{1}{\sigma_{\min}^2(\mathcal{A}_{:,J_k,:})} \left(1 - \eta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}^\top))}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:, \bar{U},:}\|_F^2}\right) \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2.
\end{aligned}$$

To ease the notation, let  $\alpha = 1 - \theta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}_{:,J,:}))}{\sigma_{\max}^2(\mathcal{A}_{:,J,:})}$ ,  $\beta = 1 - \eta \frac{\sigma_{\min}^2(\text{bcirc}(\mathcal{A}^\top))}{\|\mathcal{A}\|_F^2 - \|\mathcal{A}_{:, \bar{U}, :}\|_F^2}$ ,  $\gamma = \frac{1}{\sigma_{\min}^2(\mathcal{A}_{:,J_k,:})}$ . Then, induction over  $k$  yields

$$\begin{aligned}
\|\mathcal{X}^{(k+1)} - \mathcal{X}_*\|_F^2 &\leq \alpha \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 + \frac{1}{\gamma} \beta \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2 \\
&\leq \alpha \left( \alpha \|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2 + \frac{1}{\gamma} \beta \|\mathcal{Z}^{(k)} - \mathcal{B}^\perp\|_F^2 \right) + \beta^2 \frac{1}{\gamma} \|\mathcal{Z}^{(k-1)} - \mathcal{B}^\perp\|_F^2 \\
&\vdots \\
&\leq \alpha^{k+1} \|\mathcal{X}^{(0)} - \mathcal{X}_*\|_F^2 + \frac{1}{\gamma} \sum_{l=0}^k \alpha^l \beta^{k-l+1} \|\mathcal{Z}^{(k-1)} - \mathcal{B}^\perp\|_F^2 \\
&\leq \alpha^{k+1} \|\mathcal{X}^{(0)} - \mathcal{X}_*\|_F^2 + \frac{\alpha^{k+1} - \beta^{k+1}}{\alpha - \beta} \frac{\beta}{\alpha} \|\mathcal{B}_{R(\mathcal{A})}\|_F^2
\end{aligned}$$

# Tensor randomized greedy double block extended

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**Algorithm 1:** Tensor Random Greedy Double Block Extended Kaczmarz (TRGDBEK)

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**Input:**  $\mathcal{A} \in \mathbb{R}^{N_1 \times N_2 \times N_3}$ ,  $\mathcal{B} \in \mathbb{R}^{N_1 \times K \times N_3}$ , iters  $\ell$ ,  $\eta \in (0, 1]$ .

**Init:**  $\mathcal{X}^{(0)} = \mathcal{O}$ ,  $\mathcal{Z}^{(0)} = \mathcal{B}$ .

**for**  $k = 0, 1, \dots, \ell - 1$  **do**

$$\varepsilon_k^z = \eta \max_{j \in [N_2]} \frac{\|(\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^{(k)}\|_F^2}{\|\mathcal{A}_{:,j,:}\|_F^2}$$

$$U_k = \left\{ j \in [N_2] : \|(\mathcal{A}_{:,j,:})^\top * \mathcal{Z}^{(k)}\|_F^2 \geq \varepsilon_k^z \|\mathcal{A}_{:,j,:}\|_F^2 \right\}$$

$$\mathcal{Z}^{(k+1)} = \mathcal{Z}^{(k)} - \mathcal{A}_{:,U_k,:} * (\mathcal{A}_{:,U_k,:})^\dagger * \mathcal{Z}^{(k)}$$

Let  $J_k = [|U_k|]$  and sample  $|J_k|$  row slices with proba.  $\|\mathcal{A}_{U_k,::}\|_F^2 / \|\mathcal{A}\|_F^2$ .

$$\mathcal{X}^{(k+1)} = \mathcal{X}^{(k)} + (\mathcal{A}_{J_k,::})^\dagger * \left( \mathcal{B}_{J_k,::} - \mathcal{Z}_{J_k,::}^{(k+1)} - \mathcal{A}_{J_k,::} * \mathcal{X}^{(k)} \right)$$

**end**

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- ▶ We introduce the randomize in the TGDBEK to obtain TRGDBEK.
- ▶ In fact, TRGDBEK keeps the same update schema and the greedy ingredient, which it restricts only on the columns to force a faster convergence to the kernel.
- ▶ What changes now is: it samples randomly a block of row slices of size  $|U_k|$  provided that it is not more than the  $N_1$ .
- ▶ This allows to dynamically selects blocks of slicces, without assuming any prior partition of the rows like does TREGKB in [?].
- ▶ The random variant is well designed for large overdetermined inconsistent systems. The same idea can also be applied to Underdetermined systems.

## Numerical experiments

For the equation (??), the tensor  $\mathcal{A}$  is randomly generated using PyTorch, and the right-hand side  $\mathcal{B}$  is set to be  $\mathcal{B} = \mathcal{A} * \mathcal{X}_* + \mathcal{R} + \epsilon$ , where  $\mathcal{X}_*$  is the least-square solution of (??),  $\mathcal{R} \in \mathbb{R}(\mathcal{A}^\top)$  and  $\epsilon$  a noise term. The initial guess is given  $\mathcal{X}^{(0)} = \mathcal{O}$  and  $\mathcal{Z}^{(0)} = \mathcal{B}$ . All the computations stop when the relative error ( $err_k$ ) at the  $k$ -th iteration satisfies  $err_k < 10^{-6}$  or when  $IT$  exceeds 20000 steps. The relative error of the solution is represented by the following

$$err_k = \frac{\|\mathcal{X}^{(k)} - \mathcal{X}_*\|_F^2}{\|\mathcal{X}_*\|_F^2}$$

where  $\mathcal{X}_* = \mathcal{A}^\dagger * \mathcal{B}$  is generated using PyTorch.

# References